

Week 4 Assignment

Title: Azure Cloud Fundamentals and Data Pipeline

objective : To build an automated, end-to-end data pipeline using Azure Cloud service

Architecture Components:

Resource Group: Acts as a central container to organize all project resources.

Azure Blob Storage: Acts as the data lake. It contains an input container for the raw dataset and an output container for the final processed data.

Azure Data Factory (ADF): The main ETL (Extract, Transform, Load) engine. It uses a Linked Service to securely connect to Blob Storage.

Security (IAM): Uses Managed Identities (Storage Blob Data Contributor role) to grant ADF secure, keyless access to the storage account.

Task 1: Azure Portal & Resource Group

- Logged into the Microsoft Azure Portal.
- Searched for and opened "Resource Groups".
- Clicked on the "Create" button.
- Gave the resource group a simple name: SuperStore_Pipeline.
- Chose the region as "Central India".
- Clicked create and successfully made the resource group to hold my project.

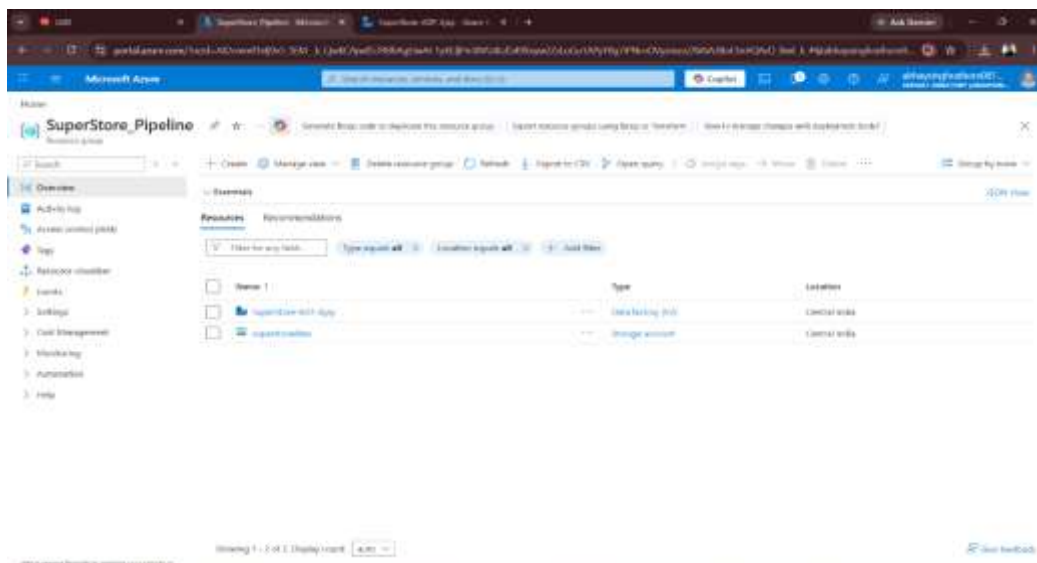


Fig 1: Resource Group Creation

Task 2: Storage Setup

- Searched for "Storage accounts" and clicked "Create".
- Gave the storage account a name and chose basic settings (Standard, LRS).
- Opened the new storage account and went to "Containers" (which act like folders).
- Created a new container named raw-data.
- Uploaded the Sample - Superstore.csv file into this container.

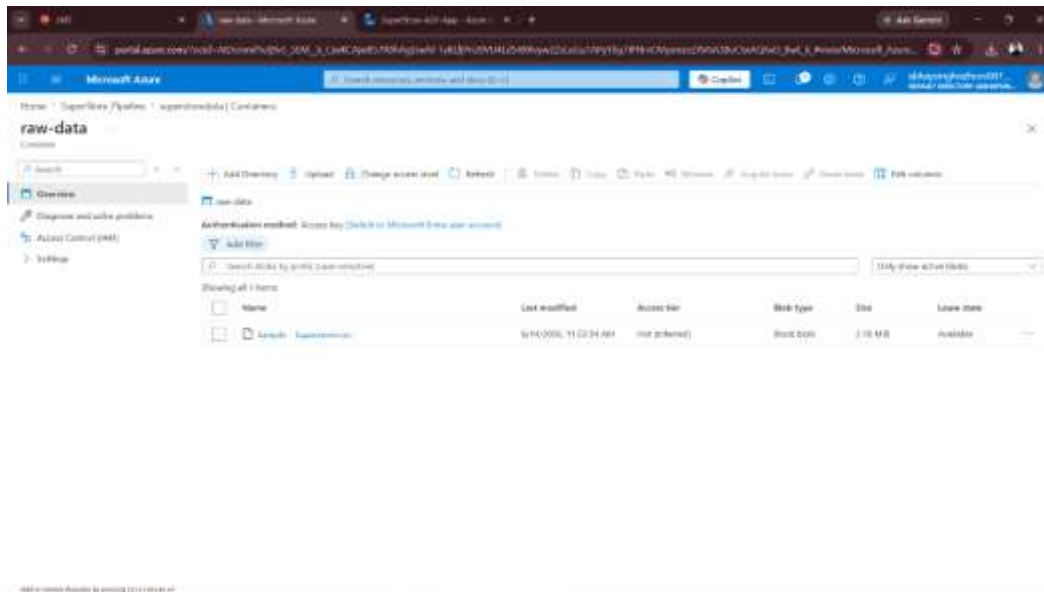


Fig 2: Storage Container with the Uploaded CSV file

Task 3: Azure Data Factory Setup

- Searched for "Data factories" in the Azure portal.
- Clicked "Create" and named my data factory SuperStore-ADF-Ajay.
- Waited for it to be ready, then clicked "Launch studio" to open my workspace

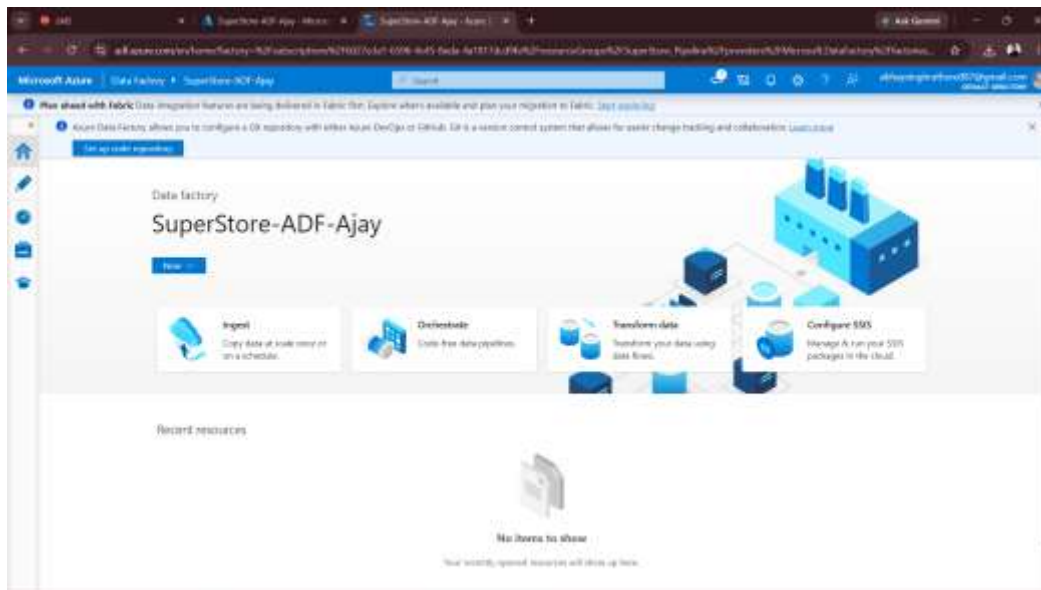


Fig 3: ADF Studio Home Screen

Task 4: Connecting the Data

- Went to the "Manage" tab in Data Factory to set up a connection.
- Created a new Linked Service named AzureBlobStorageSuperStore to securely connect my storage account.
- Went to the "Author" tab and created two simple datasets: one for my input CSV file and one for my output folder.

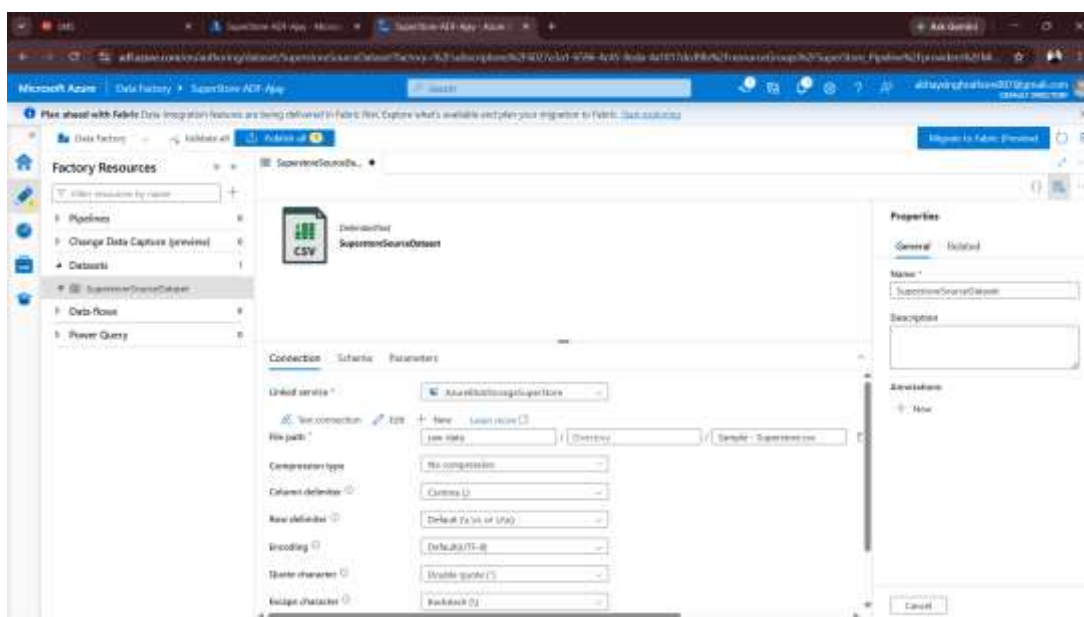


Fig 4: Linked Service and Datasets

Task 5: Building the Pipeline

- Created a new pipeline and named it SuperstoreDataPipeline.
- Added a "Get Metadata" activity to check if my source file exists.
- Added a "Copy data" activity to move the file to its new destination.
- Connected the two activities so they run one after the other.

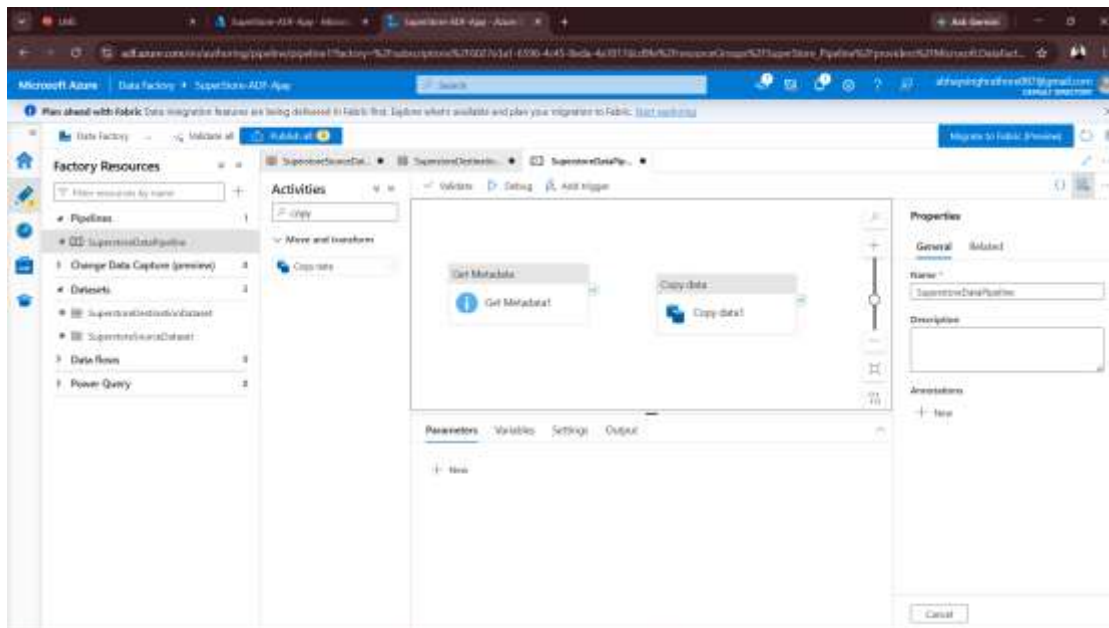


Fig 5: Pipeline Canvas

Task 6: Running the Pipeline

- Clicked the "Debug" button to run the pipeline.
- Watched the bottom panel and saw green checkmarks, meaning it was successful.
- Went back to the storage account and opened the output folder to verify my new file was copied there safely.

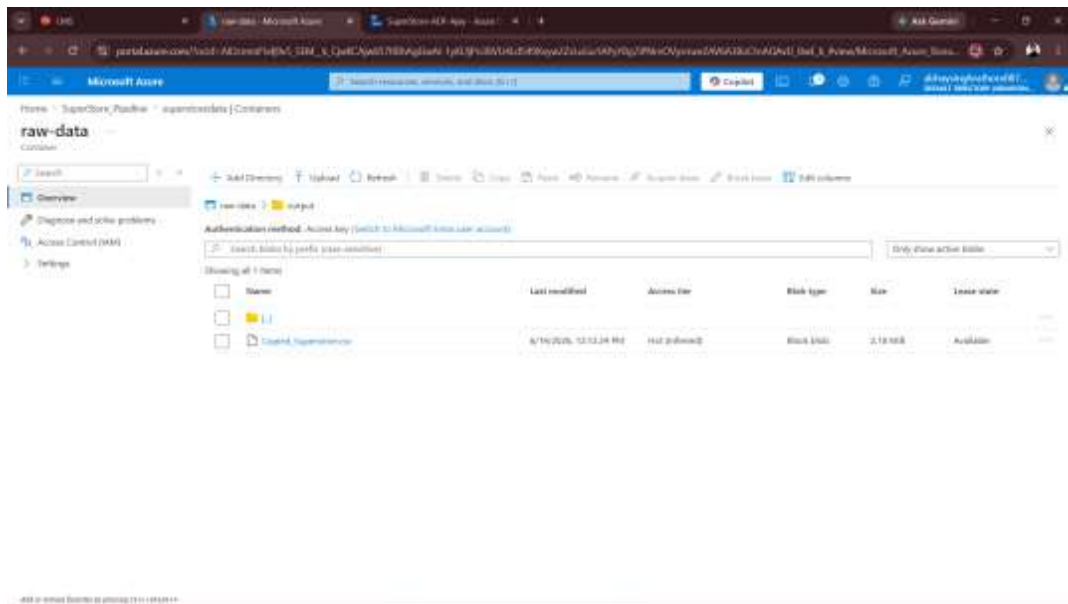


Fig 6: Pipeline Execution Succeeded

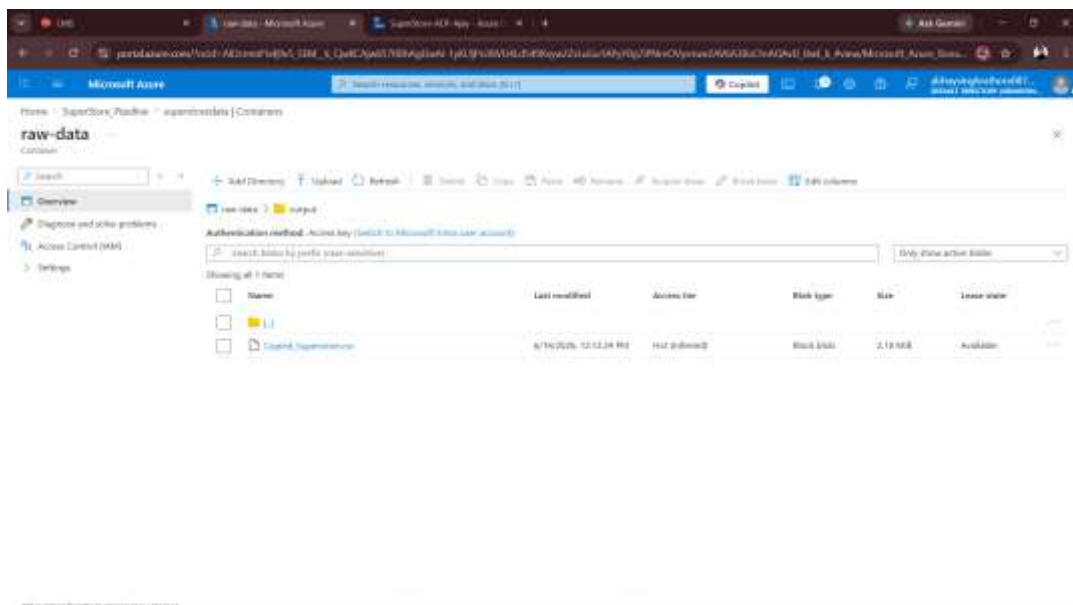


Fig 7: Destination Output File

Task 7: Security Setup (IAM)

- Opened "Access Control (IAM)" in my storage account.
- Assigned the "Storage Blob Data Contributor" role to my Data Factory.
- This gives my Data Factory secure, official permission to read and write data without using simple passwords.

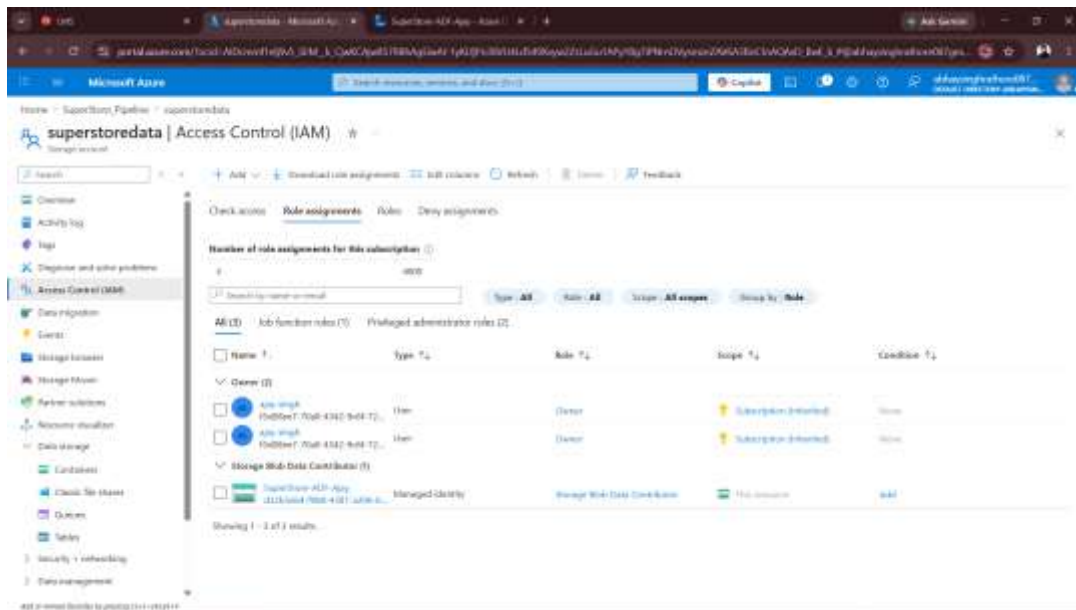


Fig 8: IAM Role Assignment